



Sergio Fuentes Cámara

IA Software Engineer - DevOps & Edge Computing Specialist

AI Software Engineer with over 5 years' experience specialising in real-time systems for edge devices. Experience implementing detection, OCR and tracking models in production, processing over 20 FPS in Full HD. Stack: Python, C++, PyTorch, DevOps, Azure.

Contact

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Technical Skills

Computer Vision & AI:

- TensorFlow & PyTorch
- Object detection & Tracking
- OCR
- Model Optimization

Edge Computing & Hardware:

- NPU acceleration
- Low-latency systems
- Embedded C/C++
- Microcontrollers & IoT

Backend & APIs:

- Python
- FastAPI & REST APIs
- WebSockets & WebRTC
- MongoDB & SQL
- Asyncio & multiprocessing

DevOps & Cloud:

- Docker
- Ansible
- CI/CD (Git, Jenkins)
- Azure (AWS & GCP)
- Linux & Bash

Frontend & Design:

- JavaScript & HTML/CSS
- React
- Flutter
- SolidWorks (3D design)
- 3D Printing (FDM/Resin)

Idiomas

Spanish : Native

English : Technical

*(documentation, papers, code)

Professional Experience

Senior Computer Vision Engineer | Grupo Etra 2023 - Present

- Architected and implemented end-to-end Computer Vision pipeline in C++ and Python processing Full HD video in real-time (<50ms latency) on NPU edge devices, reducing hardware costs by 70% vs. GPU solutions
- Designed and optimised custom object detection and OCR models achieving 97% accuracy in production, processing +15K operations/day through fine-tuning and transfer learning
- Led DevOps infrastructure for fleet of +20 edge devices implementing CI/CD with Docker, Ansible, and centralized monitoring, reducing deployment time
- Developed full-stack web platform (FastAPI, WebSocket, WebRTC) managing +10 simultaneous streams in real time, reducing remote configuration time by 5x

Junior Software Engineer | Capgemini España SL 2022 - 2023

- Developed Computer Vision system for industrial quality control automating metric measurement in production line, improving precision and reducing inspection times
- Implemented CI/CD pipelines with DevOps methodology (Git, Docker, Jenkins) for aeronautical project, ensuring continuous delivery and reducing production errors
- Designed scalable and modular software architectures following SOLID principles and design patterns, facilitating code maintainability and extensibility
- Built data preprocessing pipelines with NumPy and Pandas to feed ML models, optimizing performance and prediction quality

Electronics and Firmware Engineer | Digitanimal 2020 - 2022

- Designed and programmed low-power microcontroller firmware in C for agricultural IoT products, optimizing battery autonomy and GSM, LoRa, and Sigfox communication protocols
- Developed sensor processing algorithms (GPS, accelerometer, temperature) and efficient data transmission for real-time livestock monitoring systems
- Led full prototyping cycle from mechanical 3D design (SolidWorks) to manufacturing with 3D printing (FDM/resin), rapidly iterating to final product
- Contributed to R&D of new IoT products evaluating technical feasibility, developing POCs, and accelerating time-to-market for hardware-software innovations

Education

Bachelor's Degree in Industrial Electronics and Automation Engineering

University of Castilla-La Mancha (UCLM), Toledo
Specialization in Advanced Electronic Technologies